WORKED SOLUTION

STEP I 2016, Question 13

Probability and Statistics • Exponential Order Statistics

STEP I 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q13

The key idea

For the first and second arrivals, ask when they *haven't* happened yet. The first e-mail arrives after time t exactly when all n do, and the second arrives after t exactly when at most one has arrived by then. Both events are easy to count using independence, and differentiating the survival probability gives the density.

Solution

Let $X_1, \ldots, X_n$ be the arrival times. They're independent, and each has

$$\mathrm{P}(X_i > t) = \int_t^\infty \lambda \mathrm{e}^{-\lambda s}\,\mathrm{d}s = \mathrm{e}^{-\lambda t}, \qquad \mathrm{P}(X_i \le t) = 1 - \mathrm{e}^{-\lambda t}$$

for $t \ge 0$. We'll use the standard integral $\int_0^\infty t\,k\mathrm{e}^{-kt}\,\mathrm{d}t = \frac{1}{k}$ for $k > 0$, which is the mean of an exponential distribution with parameter k. (Integrate by parts: $[-t\mathrm{e}^{-kt}]_0^\infty + \int_0^\infty \mathrm{e}^{-kt}\,\mathrm{d}t = 0 + \frac{1}{k}$.)

Part (i)

The first arrival happens later than t if and only if every e-mail arrives later than t. By independence,

$$\mathrm{P}(T > t) = \prod_{i=1}^{n} \mathrm{P}(X_i > t) = \left(\mathrm{e}^{-\lambda t}\right)^n = \mathrm{e}^{-n\lambda t}.$$

So the cumulative distribution function is $F(t) = 1 - \mathrm{e}^{-n\lambda t}$ for $t \ge 0$. Differentiating gives the density

$$f(t) = F'(t) = n\lambda\,\mathrm{e}^{-n\lambda t} \qquad (t \ge 0),$$

as required. This is an exponential density with parameter $n\lambda$, so by the standard integral with $k = n\lambda$,

$$\mathrm{E}(T) = \frac{1}{n\lambda}.$$

Part (ii)

Let S be the arrival time of the second e-mail (so we need $n \geq 2$). The second arrival comes later than t if and only if *at most one* e-mail has arrived by time t: either none has, or exactly one has.

Ask yourself How many ways can exactly one e-mail have arrived by time t?

Any one of the n e-mails could be the early one. That one has arrived (probability $1 - \mathrm{e}^{-\lambda t}$) and the other $n-1$ haven't (probability $\mathrm{e}^{-\lambda t}$ each). By independence, and since the n cases are mutually exclusive,

$$\mathrm{P}(S > t) = \mathrm{e}^{-n\lambda t} + n\left(1 - \mathrm{e}^{-\lambda t}\right)\mathrm{e}^{-(n-1)\lambda t}.$$

Expanding the second term gives $n\mathrm{e}^{-(n-1)\lambda t} - n\mathrm{e}^{-n\lambda t}$, so

$$\mathrm{P}(S > t) = n\,\mathrm{e}^{-(n-1)\lambda t} - (n-1)\,\mathrm{e}^{-n\lambda t}.$$

As a quick sanity check, this equals 1 at $t = 0$ and tends to 0 as $t \to \infty$.

The density g of S is the derivative of $1 - \mathrm{P}(S > t)$:

$$\begin{aligned} g(t) &= -\frac{\mathrm{d}}{\mathrm{d}t}\left(n\,\mathrm{e}^{-(n-1)\lambda t} - (n-1)\,\mathrm{e}^{-n\lambda t}\right) \\ &= n(n-1)\lambda\,\mathrm{e}^{-(n-1)\lambda t} - n(n-1)\lambda\,\mathrm{e}^{-n\lambda t} \\ &= n(n-1)\lambda\left(\mathrm{e}^{-(n-1)\lambda t} - \mathrm{e}^{-n\lambda t}\right) \qquad (t \geq 0). \end{aligned}$$

For the expectation, write g as a combination of exponential densities:

$$g(t) = n \cdot (n-1)\lambda\,\mathrm{e}^{-(n-1)\lambda t} \;-\; (n-1)\cdot n\lambda\,\mathrm{e}^{-n\lambda t}.$$

Using the standard integral with $k = (n-1)\lambda$ and $k = n\lambda$ (valid because $n \geq 2$, so $k > 0$ in both cases),

$$\begin{aligned} \mathrm{E}(S) = \int_0^\infty t\,g(t)\,\mathrm{d}t &= \frac{n}{(n-1)\lambda} - \frac{n-1}{n\lambda} \\ &= \frac{1}{\lambda}\cdot\frac{n^2 - (n-1)^2}{n(n-1)} = \frac{1}{\lambda}\cdot\frac{2n-1}{n(n-1)}. \end{aligned}$$

Since $\dfrac{1}{n-1} + \dfrac{1}{n} = \dfrac{n + (n-1)}{n(n-1)} = \dfrac{2n-1}{n(n-1)}$, this is

$$\mathrm{E}(S) = \frac{1}{\lambda}\left(\frac{1}{n-1} + \frac{1}{n}\right),$$

which is the required result.

Ask yourself Why is the answer a sum of two simple terms?

The exponential distribution is memoryless. The first arrival comes after an exponential time with rate $n\lambda$, with mean $\frac{1}{n\lambda}$. After that, the remaining $n-1$ e-mails start afresh, as though the clock had just been reset, so the further wait is exponential with rate $(n-1)\lambda$ and mean $\frac{1}{(n-1)\lambda}$. Adding the two waits gives the answer at once, and it's a good way to remember it.

Checks

- With $n = 2$ and $\lambda = 1$, the second arrival is the later of two independent exponential times. Its mean should be $1 + \frac{1}{2} = 1.5$, and a computer simulation of 10^5 trials gave 1.501.
- The density g integrates to 1: $\int_0^\infty g = n - (n-1) = 1$.
- With $n = 5$ and $\lambda = 1$, the formula gives $\mathrm{E}(S) = \frac{1}{4} + \frac{1}{5} = 0.45$, which agrees with a numerical evaluation of $\int_0^\infty t\, g(t)\, \mathrm{d}t$.

Common slips

- Forgetting the factor n when exactly one e-mail has arrived. Any of the n could be the early one.
- Writing $\mathrm{P}(S > t)$ as the probability that *exactly* one has arrived, which leaves out the case where none has.

Beyond STEP

T and S are the first two *order statistics* of an exponential sample. The memoryless argument above extends to show that the kth arrival has mean $\frac{1}{\lambda}\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{n-k+1}\right)$, and this is the basis of the Poisson process and of queueing theory. Blitzstein and Hwang, *Introduction to Probability*, covers order statistics and the memoryless property clearly.

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